

Chapter 4: Project Management

4.1 Risk Identification

The possible risks of the Internship Tracker System were identified based on the system's technology, people, organizational environment, tools, requirements, and project estimation. Table 1 presents the identified risks and their possible causes.

Table 1. Risk Identification Table

Risk Type	Possible Risks
Technology Risks	The database may become slow or unstable when many students simultaneously access application records, company information, and dashboards.
	The system may have difficulty securely storing and retrieving uploaded CVs and cover letters.
	Email notifications may fail or be delayed because of problems with the external email service.
	Real-time application status updates may not be reflected correctly across the student's dashboard and company or university views.
People Risks	A developer responsible for an important part of the system, such as authentication, or post management, may become unavailable during development.
	Team members may not have enough experience to properly implement security, or document uploads.
	Poor communication between developers may cause different parts of the system to be implemented in incompatible ways.
	Developers may misunderstand requirements and implement features differently from what students, companies, or university authorities expect.
Organizational Risks	Changes in company internship policies may require changes to application statuses or required documents.
	Limited project resources may force the team to reduce or remove lower-priority features such as AI recommendations or advanced analytics.
	Companies may introduce new rules about how internship information should be stored or accessed.
	Changes in the responsibilities of companies or university authorities may require changes to their system functions and access permissions.
Tools Risks	The libraries used for authentication, file handling, or notifications may conflict with each other or with the project's software version.
	The tools used for email delivery may not integrate reliably with the Internship Tracker System.
	Development tools may become incompatible after a software or library update.
	Testing tools may not adequately simulate multiple students accessing the system at the same time.
Requirements Risks	Companies may request new application stages after development has started, requiring changes to the status-tracking functions.
	Requirements for uploaded documents may change, such as allowing new file

Risk Type	Possible Risks
	types or requiring additional documents.
	Different stakeholders may disagree about what information companies, students, and university authorities should be able to view.
	Changes in the requirements for AI-based internship recommendations may require changes to the recommendation method and user profile data.
	Stakeholders may not clearly define their requirements during elicitation, causing incorrect features to be developed.
Estimation Risks	The team may underestimate the time required to develop and test application tracking, document management, notifications, and user authentication.
	The amount of testing required for different users, application statuses, file types, and notification conditions may be underestimated.
	The team may underestimate the time required to fix integration problems between the codes, database, and email service.

4.2 Risk Analysis

After identifying the risks, their probability and possible effect on the project were assessed. The probability indicates how likely a risk is to occur, while the effect shows how seriously it could affect the project. Table 2 presents the probability and effect assigned to each identified risk.

Table 2. Risk Analysis Table

Identified Risk	Probability	Effect
The database may become slow or unstable when many students simultaneously access application records, company information, and dashboards.	Moderate	Serious
The system may have difficulty securely storing and retrieving uploaded CVs and cover letters.	Moderate	Serious
Email notifications may fail or be delayed because of problems with the external email service.	Moderate	Serious
Application status changes may not appear correctly across student, company, and university views.	Moderate	Serious
A developer responsible for an important part of the system, such as authentication or post management, may become unavailable during development.	Low	Serious
Team members may not have enough experience to properly implement security or document-upload functions.	Moderate	Serious
Poor communication between developers may cause different parts of the system to be implemented in incompatible ways.	Moderate	Tolerable
Developers may misunderstand requirements and implement features differently from what students, companies, or university authorities expect.	Moderate	Serious
Changes in company internship policies may require changes to application statuses or required documents.	Moderate	Serious
Limited project resources may force the team to reduce or remove lower-	Low	Tolerable

Identified Risk	Probability	Effect
priority features such as AI recommendations or advanced analytics.		
Companies may introduce new rules about how internship information should be stored or accessed.	Low	Serious
Changes in the responsibilities of companies or university authorities may require changes to their system functions and access permissions.	Low	Serious
The libraries used for authentication, file handling, or notifications may conflict with each other or with the project's software version.	Moderate	Serious
The tools used for email delivery may not integrate reliably with the Internship Tracker System.	Moderate	Serious
Development tools may become incompatible after a software or library update.	Low	Tolerable
Testing tools may not adequately simulate multiple students accessing the system at the same time.	Moderate	Tolerable
Companies may request new application stages after development has started, requiring changes to the status-tracking functions.	High	Serious
Requirements for uploaded documents may change, such as allowing new file types or requiring additional documents.	Moderate	Tolerable
Different stakeholders may disagree about what information companies, students, and university authorities should be able to view.	Moderate	Serious
Changes in the requirements for AI-based internship recommendations may require changes to the recommendation method and user profile data.	Moderate	Serious
Stakeholders may not clearly define their requirements during elicitation, causing incorrect features to be developed.	Moderate	Serious
The team may underestimate the time required to develop and test application tracking, document management, notifications, and user authentication.	High	Serious
The amount of testing required for different users, application statuses, file types, and notification conditions may be underestimated.	High	Serious
The team may underestimate the time required to fix integration problems between the code, database, and email service.	Moderate	Tolerable

4.3 Risk Planning

Appropriate actions were planned to reduce the likelihood or impact of the identified risks. The plans include avoidance, minimization, and contingency measures where necessary. Table 3 presents the planned response for each risk.

Table 3. Risk Planning Table

Risk	Risk Management Plan
Database becomes slow under heavy usage.	Optimize queries and add database indexes.
CVs and cover letters cannot be stored or retrieved properly.	Validate files and keep regular backups.

Risk	Risk Management Plan
Email notifications fail or are delayed.	Use a reliable email service and retry failed emails.
Application status is not updated correctly across user views.	Use one central status record and test all status changes.
A key developer becomes unavailable.	Share tasks and document important code.
Team members lack security or file-handling skills.	Provide training and conduct code reviews.
Poor communication causes incompatible code.	Hold regular meetings and use version control.
Developers misunderstand stakeholder requirements.	Review requirements with stakeholders before development.
Company internship policies change.	Make application statuses and documents easy to modify.
Limited resources force removal of lower-priority features.	Develop high-priority features first.
Companies introduce new data access or storage rules.	Keep user permissions configurable.
Company or university responsibilities change.	Keep roles and permissions separate from core functions.
Libraries conflict with the project software.	Check compatibility before installing libraries.
Email tools do not integrate properly.	Test the email service before full implementation.
Development tools become incompatible after updates.	Record software versions and test updates separately.
Testing tools cannot simulate many users.	Test critical functions separately.
Companies request new application stages.	Make application statuses easy to add or modify.
Document requirements change.	Make file validation rules easy to update.
Stakeholders disagree about information access.	Define access permissions during requirements negotiation.
Stakeholders do not clearly define requirements.	Use requirement reviews and prototypes.
Development time is underestimated.	Divide work into smaller tasks and set priorities.
Testing effort is underestimated.	Prepare a testing plan before development ends.
Integration problems take longer than expected.	Test database, code, and email integration separately.